

Aim

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Benchmarking: cell type annotation

Code ▾

2026-08-10

Runs: pilot 260423_VH01624_453_222HWMYX (HBDN206-MNpCT, HBDN392-AML-MDS, HBDN501-AML-KMT2A, MOLM13) and 260528_VH01624_464_222K7VKX / LK2 (normal-01, HBDN376-TP53, HBDN498-TP53, APOP576-TP53).

Analysis finished: 2026-08-09 - the most recent of the 37 figures shown below.

Report generated: 2026-08-10 by `05.celltype_annotation_report.Rmd`, from the outputs of steps 01-04 in this folder.

Cells are annotated by projecting them into the Zeng et al. bone marrow reference with Symphony, rather than by clustering and scoring marker genes. The reference carries its own mapping-error QC, so a cell can come back with no label at all. This report covers how the projection behaves on our data, how many cells it refuses, what those cells are, and how much the labels can be trusted.

For full report, download `celltype_annotation_report.html` in the attachment and open in a browser.

Code used - attached and in the github repository:

https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/7.Benchmarking-cell_type_annotation
(https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/7.Benchmarking-cell_type_annotation)

Aim

Establish reference-based annotation as the project's cell typing, and find out where it breaks:

- how many cells does the reference refuse to label, and is that number stable between runs
- what are the refused cells - real biology the reference does not contain, or bad cells
- how much confidence sits behind the labels that do survive, and for which populations
- fine versus broad labels: which one downstream analysis should key on

Conclusions

- **BoneMarrowMap projection works and is what the project uses.** It annotates 92.05% of pilot cells and 94.52% of LK2 cells, at the default `MAD_threshold` of 2.5
- **the cells it refuses are bad cells, not unknown biology.** They are half the depth of the cells that pass (837 vs 2556 genes in LK2), have higher mitochondrial and **four times lower ribosomal** content, and are enriched for HTO-negative droplets. Forced to pick a neighbour they land on late erythroid and monocyte, the two lowest-complexity populations
- **the rejection rate is a readout of how far a sample sits from normal marrow.** In the pilot the normal donor loses 4.8% and the leukaemias 10-11%; in LK2 APOP576-TP53 loses **30%** against 3-4% for the rest. A high reject rate is a flag to look at the sample, not at the reference
- **`predict_CellTypes()` does not re-label anything.** The final label is identical to `initial_CellType` for 100% of cells; the only thing it does is set the QC-failed cells to NA. The NA cells and the `mapping_error_QC == "Fail"` cells are the same cells, 1:1

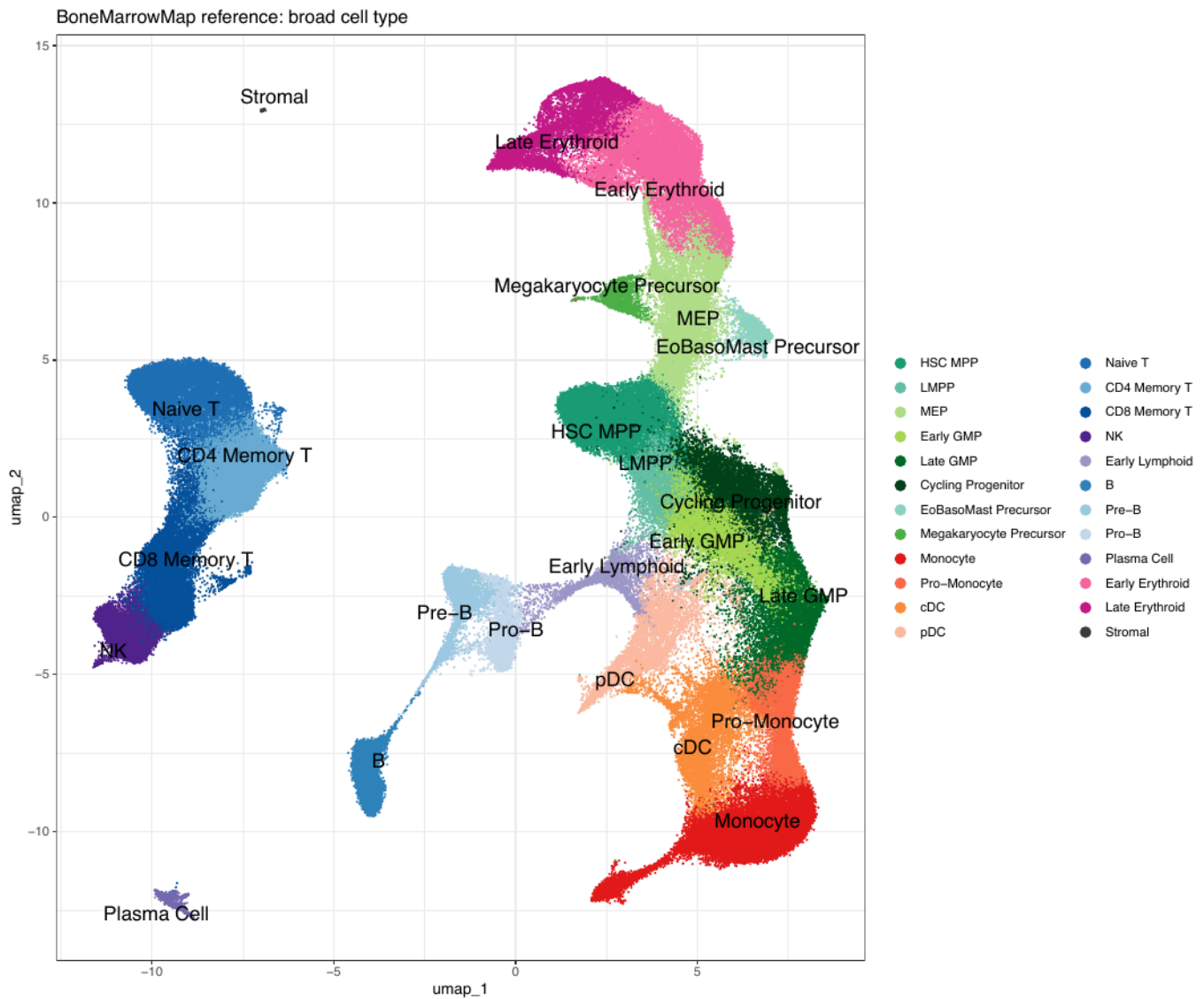
- **label confidence tracks how continuous the biology is, not how well the projection ran.** Terminal populations - Monocyte, cDC, pDC, Plasma Cell, Late Erythroid - come back at a median vote fraction of 1.0. HSC MPP, LMPP, Early Lymphoid and Cycling Progenitor sit at 0.57-0.67, because the reference is resolving a gradient there rather than a boundary
- **downstream grouping must key on `predicted_CellType_Broad`, not on `predicted_CellType`.** The fine column has 53 levels and only 9 of them appear in the broad vocabulary the lineage lists and the bespoke palette are written against. Keyed on the fine label, 84.3% of pilot cells and 89.1% of LK2 cells fall into "Other"; keyed on the broad label, none do
- **the Unknown cells sit with erythroid and pro-monocyte.** On `02a_umap_native_unknown_highlighted.pdf` the majority fall in the erythroid / late erythroid territory and in pro-monocyte, rather than scattering across the map
- **the pipeline and the standalone scripts disagree on the threshold.** `nextflow/bin/annotate.R` runs at `MAD_threshold = 4`; steps 01 and 02 ran at the BoneMarrowMap default of 2.5. Going 2.5 -> 3 -> 4 roughly halves the rejection rate (pilot 7.95 -> 6.36 -> 3.91%, LK2 5.48 -> 3.58 -> 1.75%) without changing which samples are hardest to place. Given what section 3 shows the rejected cells to be, 2.5 is the better setting - but 4 is what production currently runs, and that should be settled on purpose
- there is **no confidence cutoff anywhere in the annotation.** `predict_CellTypes()` reports the `k = 30` vote share and never thresholds on it; the only gate is the mapping-error QC

1. The reference

The reference is BoneMarrowMap (Zeng et al.), a Symphony reference built on normal human bone marrow. It ships as a 345 MB reference object plus a 221 MB uwot model, both downloaded once into `annotation/`. Projection is `map_Query()` -> `calculate_MappingError()` -> `predict_CellTypes()` -> `predict_Pseudotime()`, with the demultiplexed sample as the batch variable so Symphony harmonises across donors within a pool rather than treating the GEM well as one batch.

00a_BoneMarrowMap_reference_broad_celltype.pdf

The reference itself, coloured by broad cell type in the project palette. Query cells are placed into this space; they do not move it. Stromal is the grey island - it is in the reference, and it is the population that used to be dropped from every figure by the palette bug described in section 9.



The reference returns two label columns, and the difference between them is the single thing most likely to cause trouble downstream:

How the fine labels roll up into the broad ones (10 broadest, LK1 + LK2 pooled)

predicted_CellType_Broad	n_fine	fine_labels
CD8 Memory T	5	CD8 Effector Memory 2, CD8 Effector Memory 1, CD8 Central Memory, CD8 Tissue Resident Memory
Early Lymphoid	4	CLP, MLP, Pre-ProB, MLP-II
cDC	4	cDC2, Pre-cDC, cDC1, ASDC
CD4 Memory T	3	CD4 Central Memory, CD4 Effector Memory, CD4 Regulatory
Early Erythroid	3	CFU-E, Basophilic Erythroblast, Pro-Erythroblast
HSC MPP	3	HSC, MPP-MyLy, MPP-MkEry
Late GMP	3	GMP-Mono, GMP-Neut, GMP-Cycle
NK	3	NK, NK CD56high, NK Proliferating
pDC	3	pDC, Pre-pDC, Pre-pDC Cycling
B	2	Mature B, Immature B

2. How many cells the reference refuses

`calculate_MappingError()` scores each cell on how well it sits in reference space and calls anything beyond `MAD_threshold` median absolute deviations a Fail. Failed cells keep their projected coordinates but lose their label.

Mapping-error QC, whole run (MAD_threshold = 2.5)

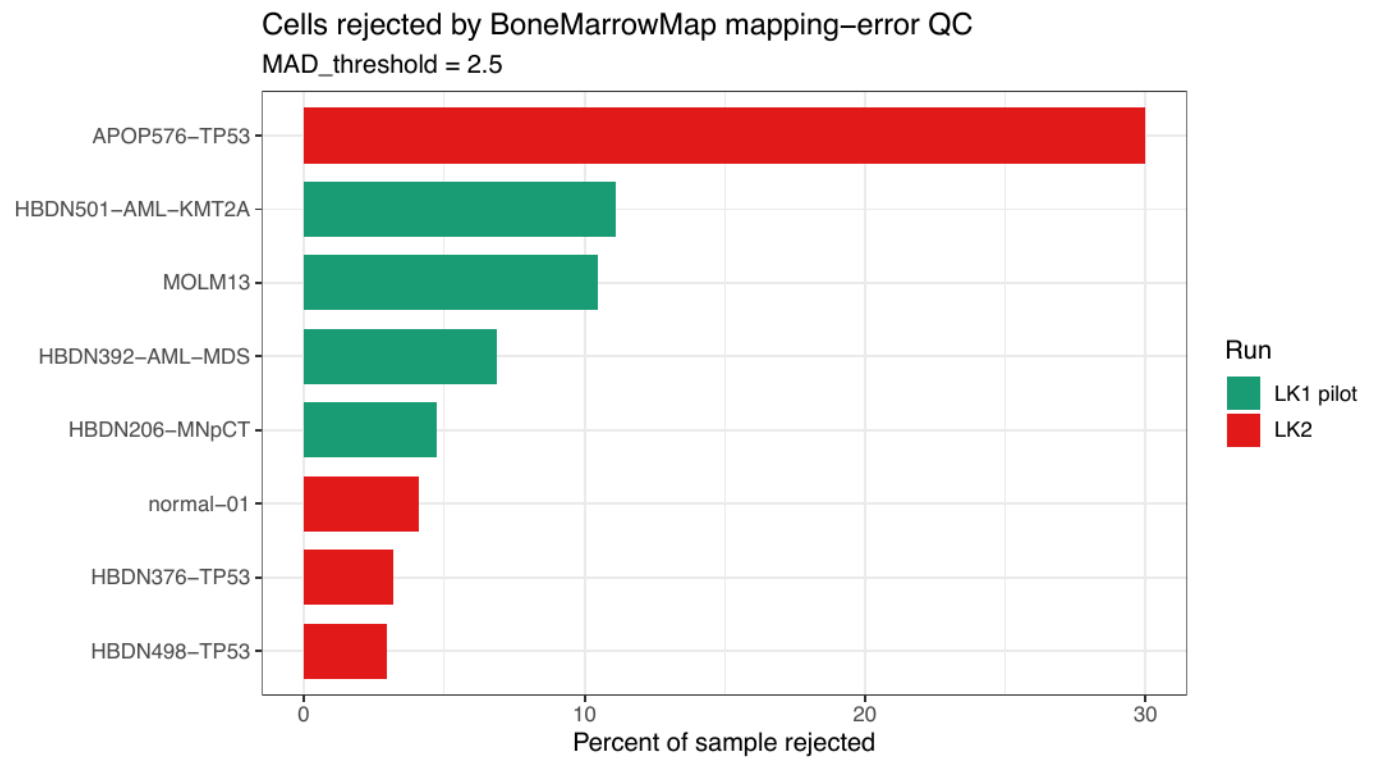
run	mapping_error_QC	n	total	percent
LK1 pilot	Fail	1713	21541	7.95
LK1 pilot	Pass	19828	21541	92.05
LK2	Fail	1195	21800	5.48
LK2	Pass	20605	21800	94.52

Rejection rate per demultiplexed sample, worst first

run	sample	mapping_error_QC	n	total	percent
LK2	APOP576-TP53	Fail	543	1812	29.97
LK1 pilot	HBDN501-AML-KMT2A	Fail	351	3154	11.13
LK1 pilot	MOLM13	Fail	656	6262	10.48
LK1 pilot	HBDN392-AML-MDS	Fail	422	6144	6.87
LK1 pilot	HBDN206-MNpCT	Fail	284	5981	4.75
LK2	normal-01	Fail	146	3568	4.09
LK2	HBDN376-TP53	Fail	287	9053	3.17
LK2	HBDN498-TP53	Fail	219	7367	2.97

01_mapping_QC_fail_by_sample.pdf

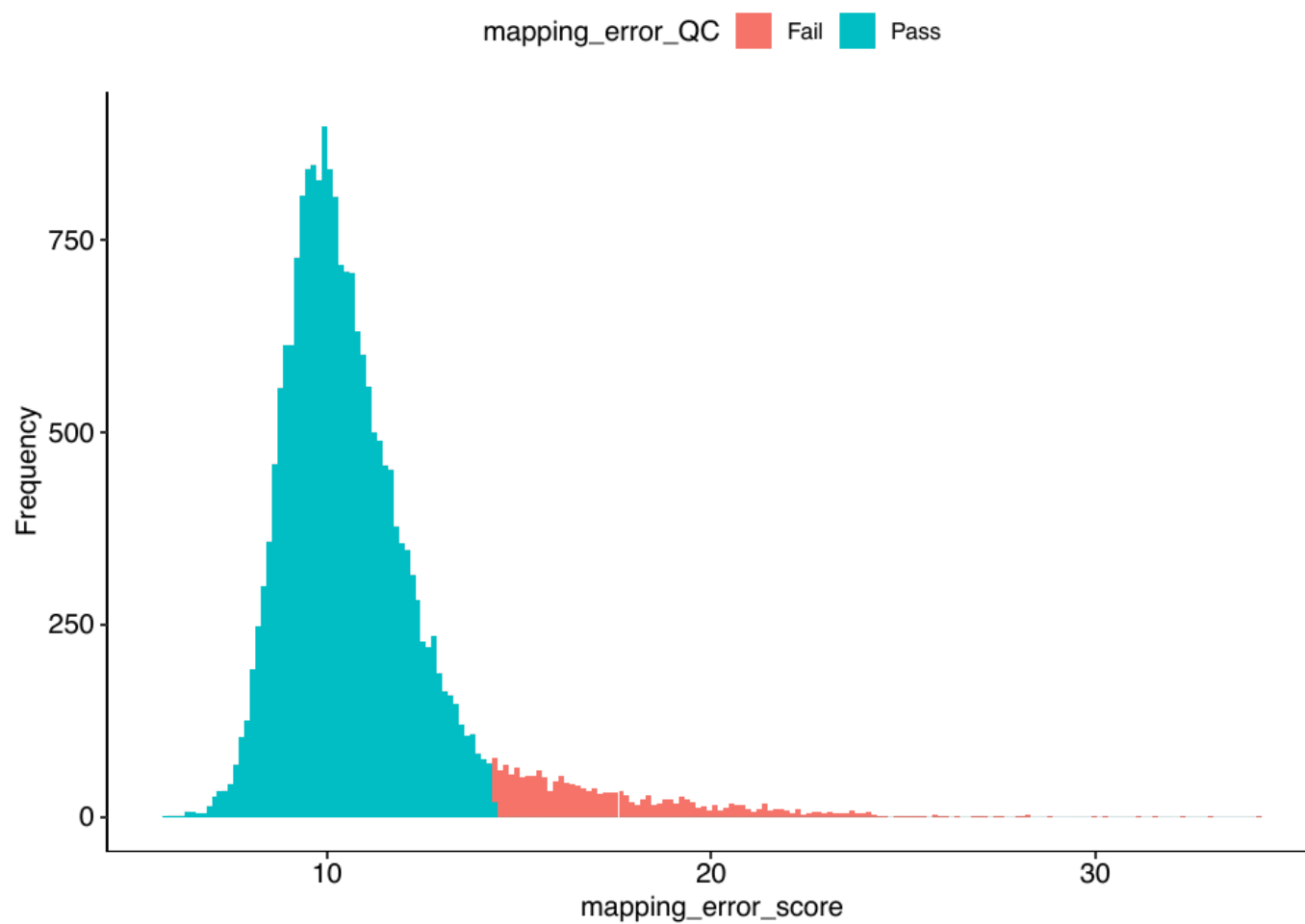
Per-sample rejection. The normal donors sit at the bottom; the leukaemias and the MOLM13 cell line sit at the top. APOP576-TP53 is the outlier at 30%.



01_mapping_error_QC.pdf

The mapping error distribution and the threshold, pilot run.

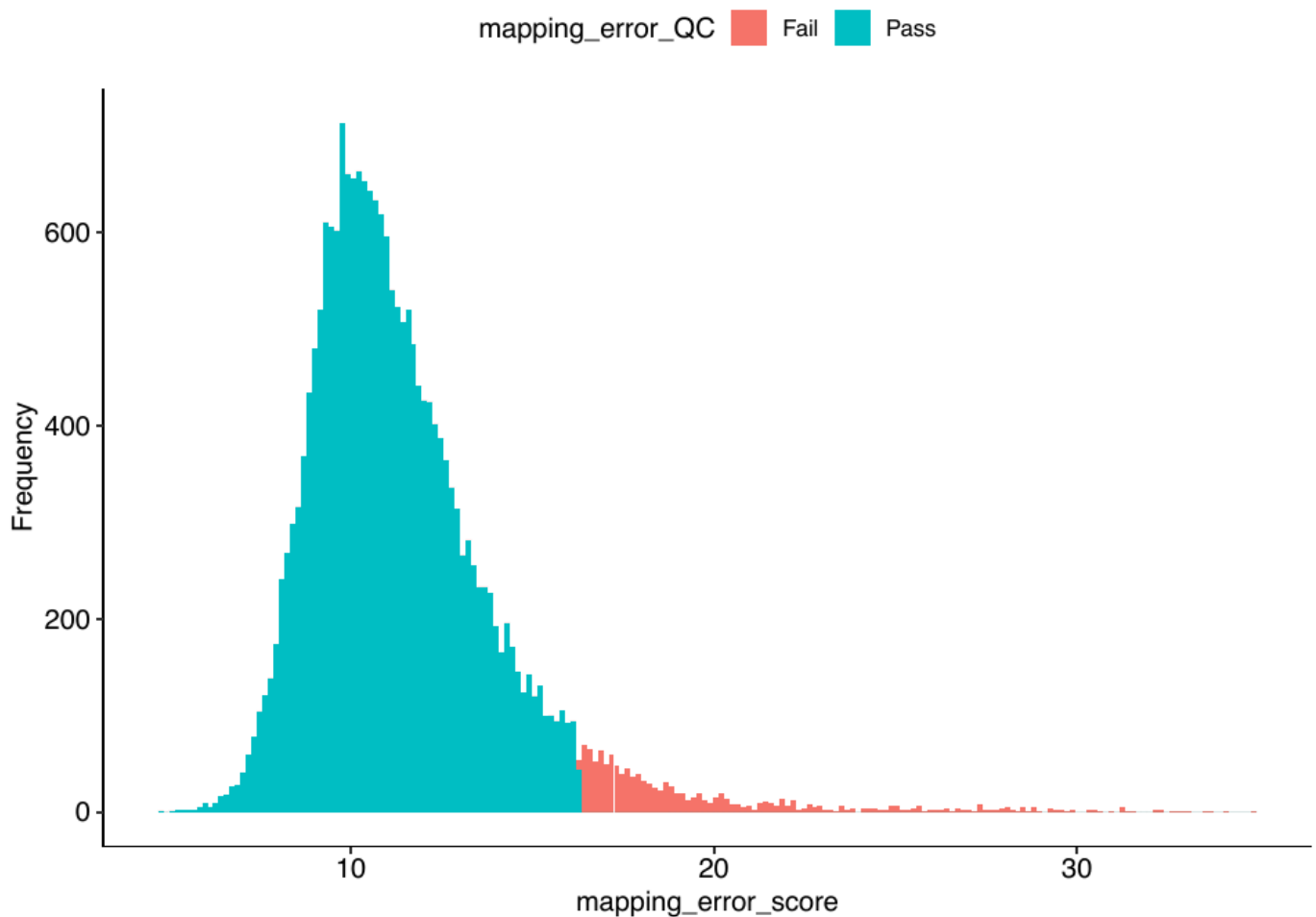
Mapping Error Cutoff



01_mapping_error_QC.pdf

The same for LK2.

Mapping Error Cutoff



The ordering is the point. A reference built on normal marrow places normal cells well and leukaemic cells less well, so the rejection rate reads as distance from normal rather than as a technical failure. It is worth checking whenever a new sample comes in.

3. What the refused cells are

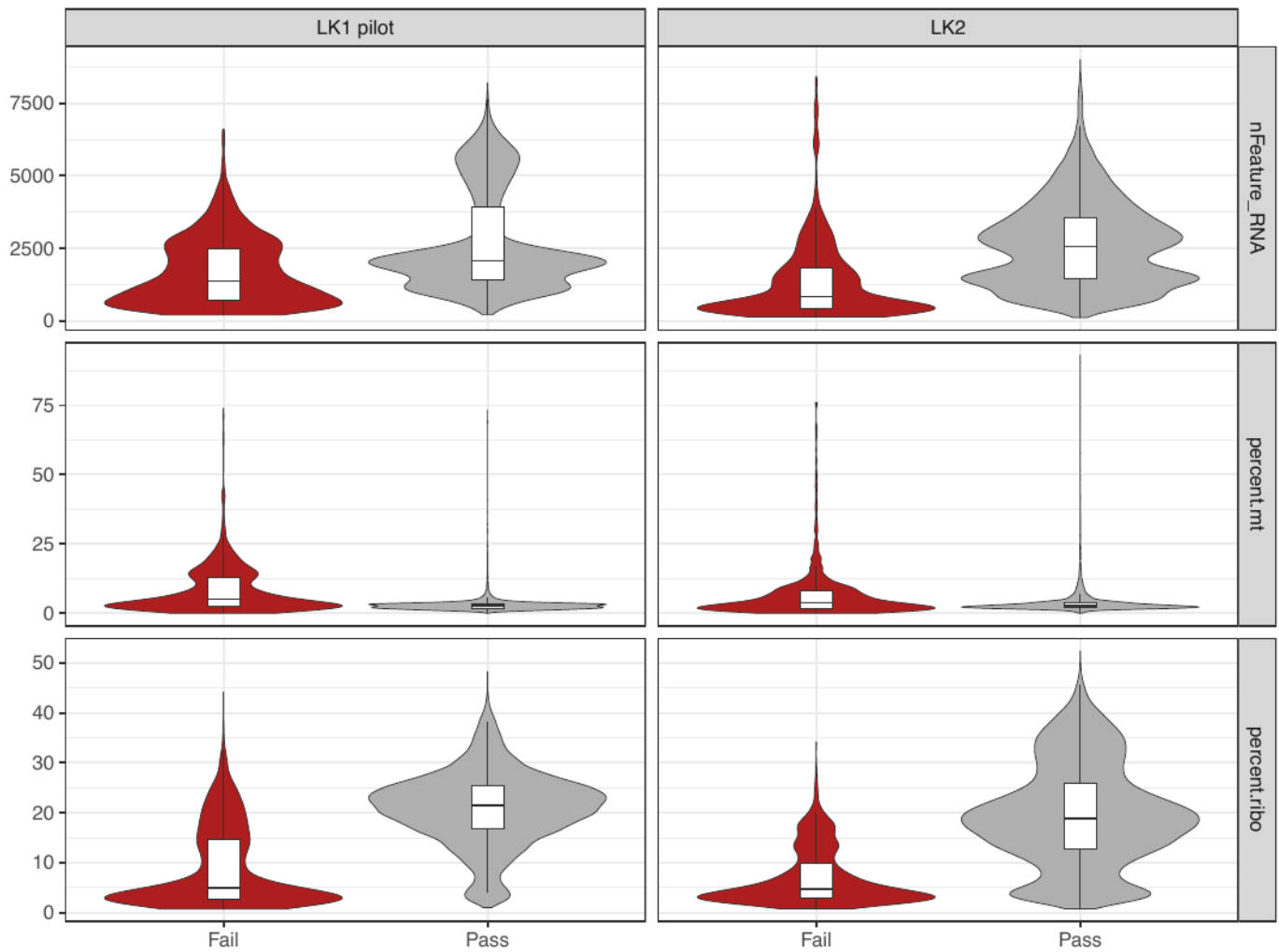
Median QC covariates, cells that pass versus cells that fail

run	mapping_error_QC	n	nCount_RNA	nFeature_RNA	percent.mt	percent.ribo	scDblFinder.score	mapping_error_score
LK1 pilot	Fail	1713	2300	1370	5.092	4.970	0.390	16.657
LK1 pilot	Pass	19828	4272	2076	2.646	21.424	0.394	10.199
LK2	Fail	1195	1380	837	3.910	4.657	0.063	18.081
LK2	Pass	20605	6005	2556	2.776	18.804	0.150	10.767

02_rejected_cell_profile.pdf

The rejected cells are shallower, more mitochondrial and much less ribosomal than the cells that pass. The ribosomal drop is the clearest separator - roughly a four-fold difference in both runs.

QC covariates of cells the reference rejects



Step 03 pushes this further on the LK2 pipeline checkpoint, where the same projection is run at `MAD_threshold = 4` and the failed cells are followed individually.

Unknown versus annotated cells, median QC covariates (`MAD_threshold = 4`)

is_unknown	n	nCount_RNA	nFeature_RNA	percent.mt	percent.ribo	scDblFinder.score
FALSE	25351	6211.0	2578.0	2.762967	19.757907	0.0003023
TRUE	346	772.5	511.5	3.225688	6.270695	0.0000054

Unknown fraction by donor

sample_name	is_unknown	n	percent
APOP576-TP53	TRUE	182	7.9372002
normal-01	TRUE	68	1.7046879
HBDN376-TP53	TRUE	63	0.6673729
HBDN498-TP53	TRUE	33	0.3308271

HTO classification of Unknown versus annotated cells

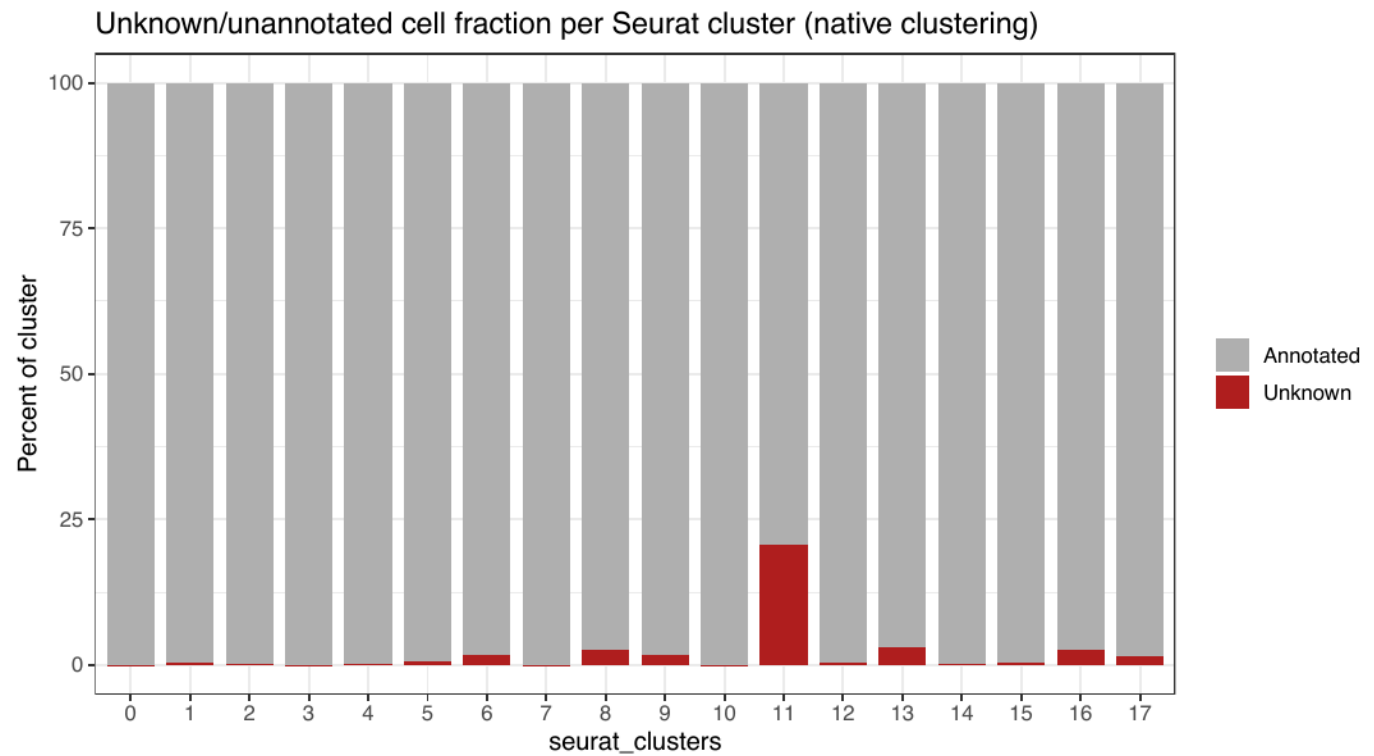
is_unknown	HTO_classification.global	n	percent
FALSE	Doublet	2946	11.620843
FALSE	Negative	2503	9.873378
FALSE	Singlet	19902	78.505779

is_unknown	HTO_classification.global	n	percent
TRUE	Doublet	53	15.317919
TRUE	Negative	74	21.387283
TRUE	Singlet	219	63.294798

Unknown cells are enriched in HTO-negative droplets (21.4% against 9.9%), which is the same population that fails hashing - low-content droplets that carry neither enough transcriptome to place nor enough hashtag to assign.

01_unknown_fraction_by_cluster.pdf

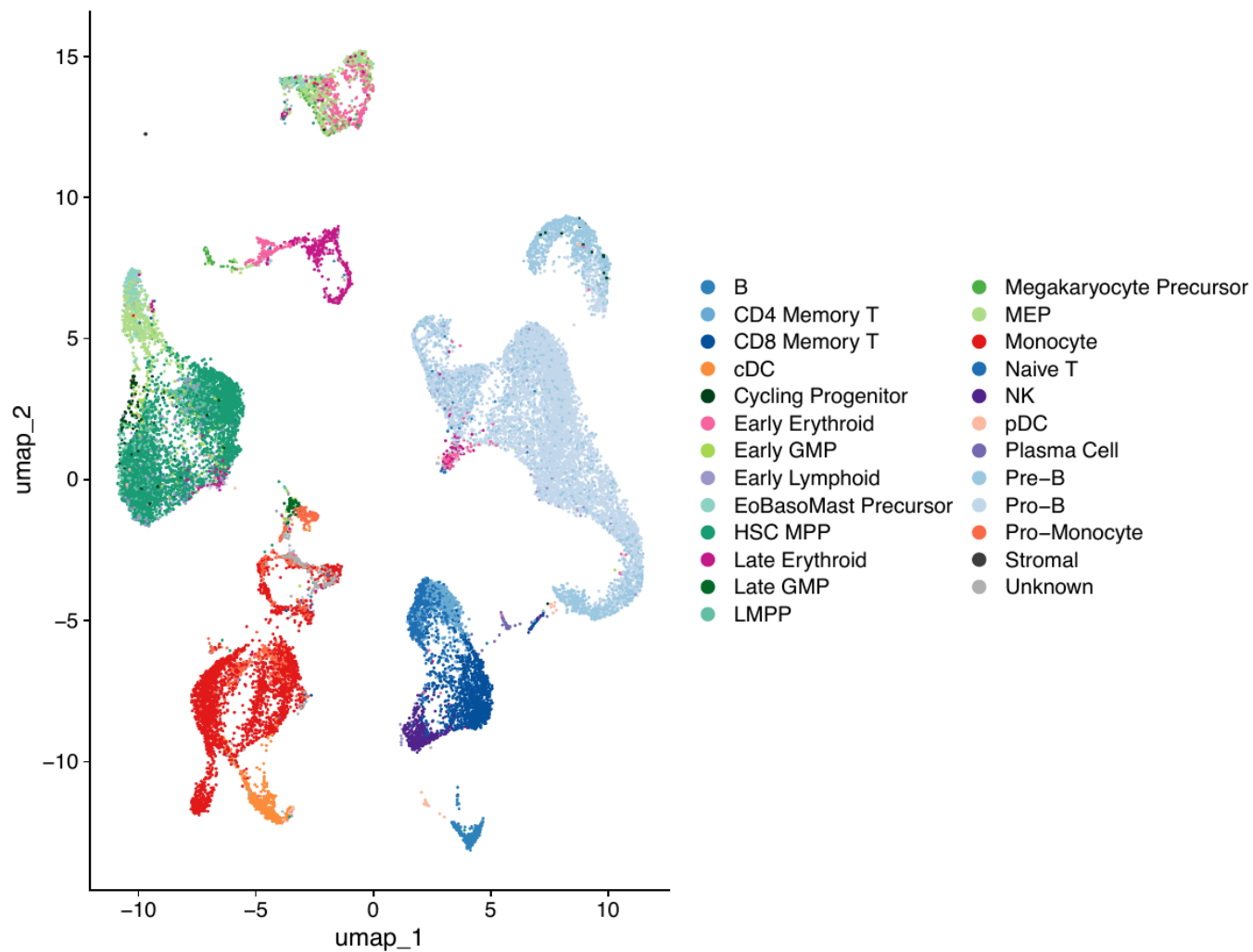
Unknown cells are not spread evenly: one native cluster (11) is a fifth Unknown, the rest are under 3%. They form a compartment, not a background.



02_umap_native_with_unknown.pdf

Native UMAP with Unknown as its own category.

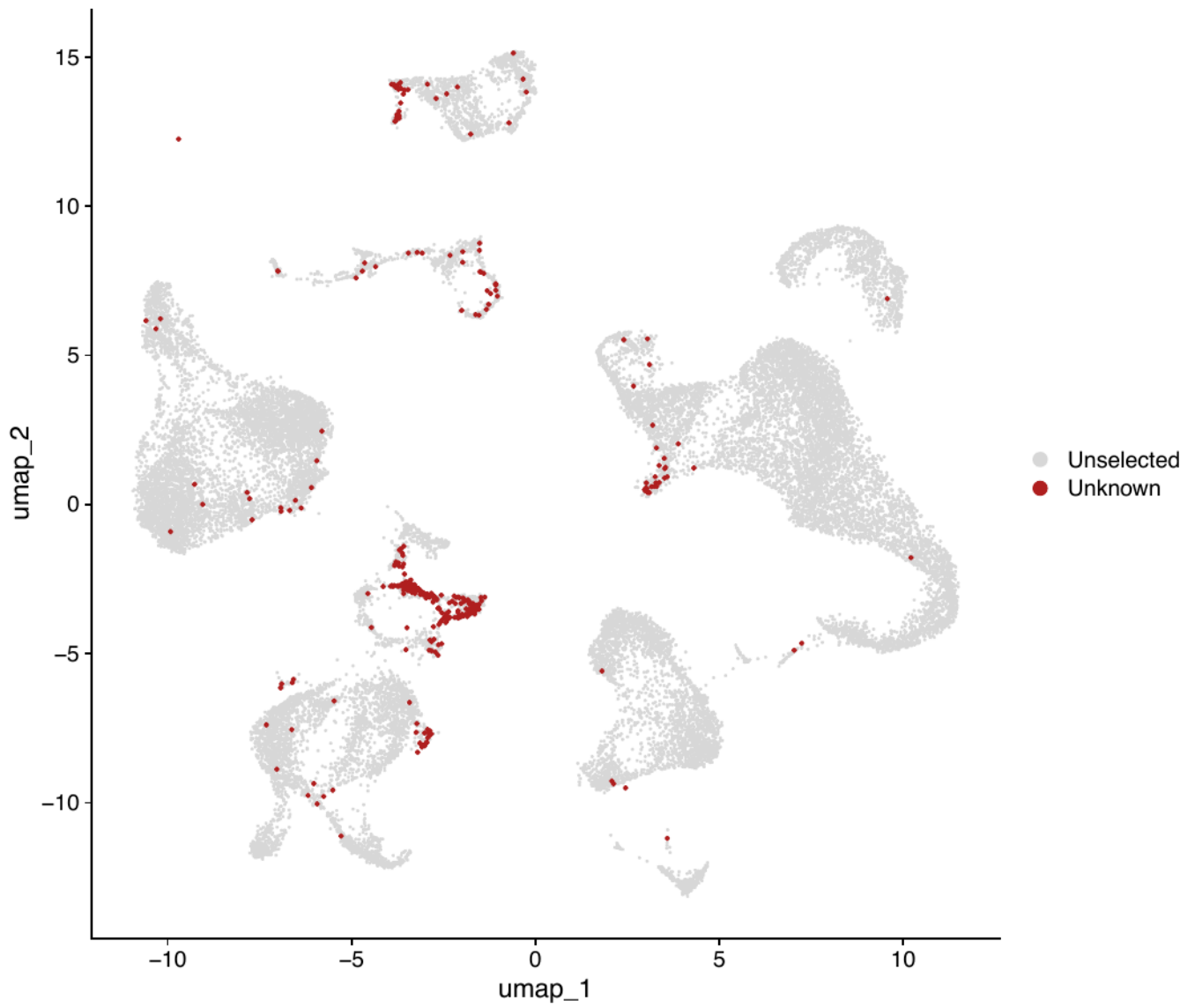
iot-projected) UMAP – BoneMarrowMap annotation incl. Unknown



02a_umap_native_unknown_highlighted.pdf

The same, with only the Unknown cells picked out - they sit together rather than scattering across the manifold.

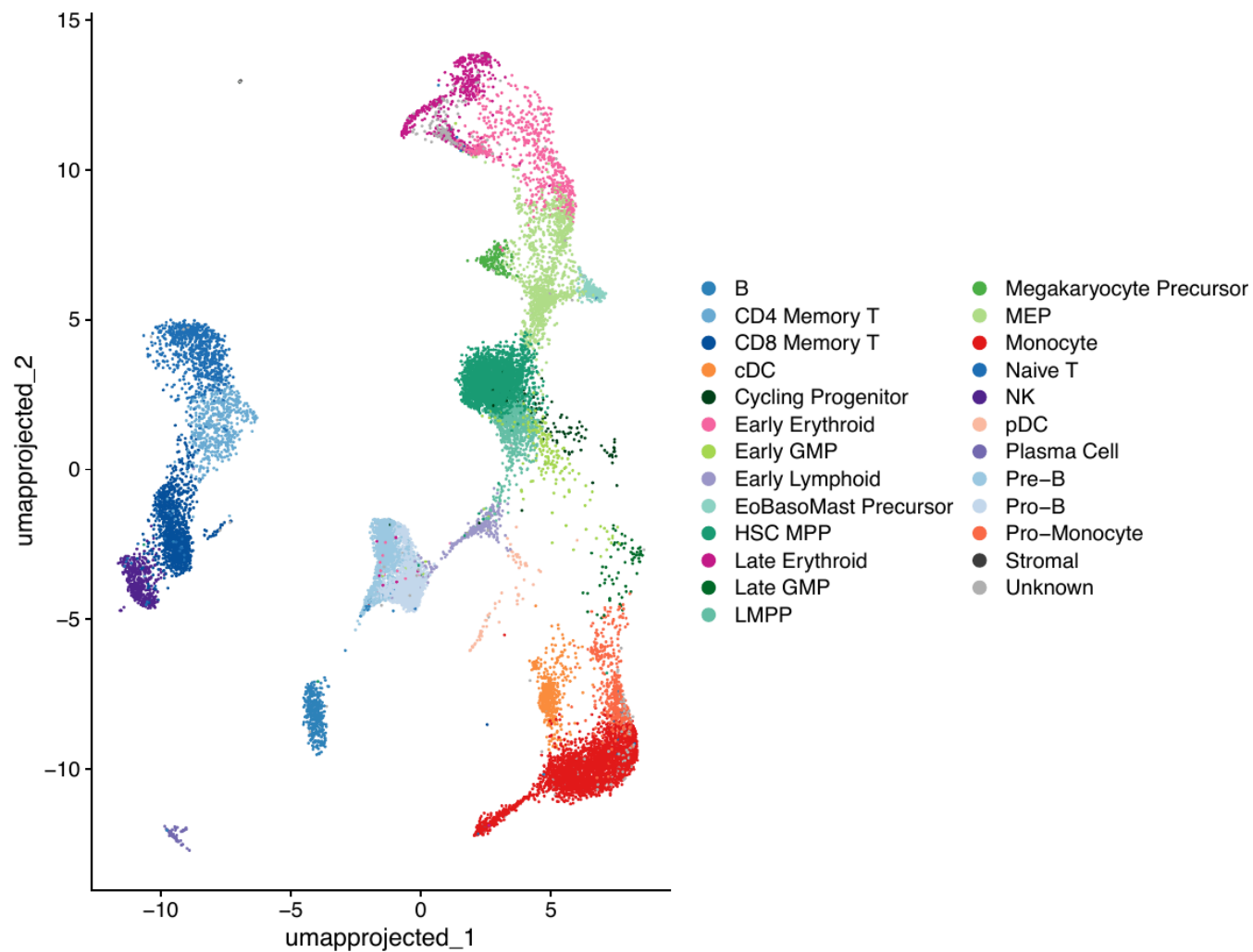
Native UMAP – Unknown cells highlighted



03_umap_projected_with_unknown.pdf

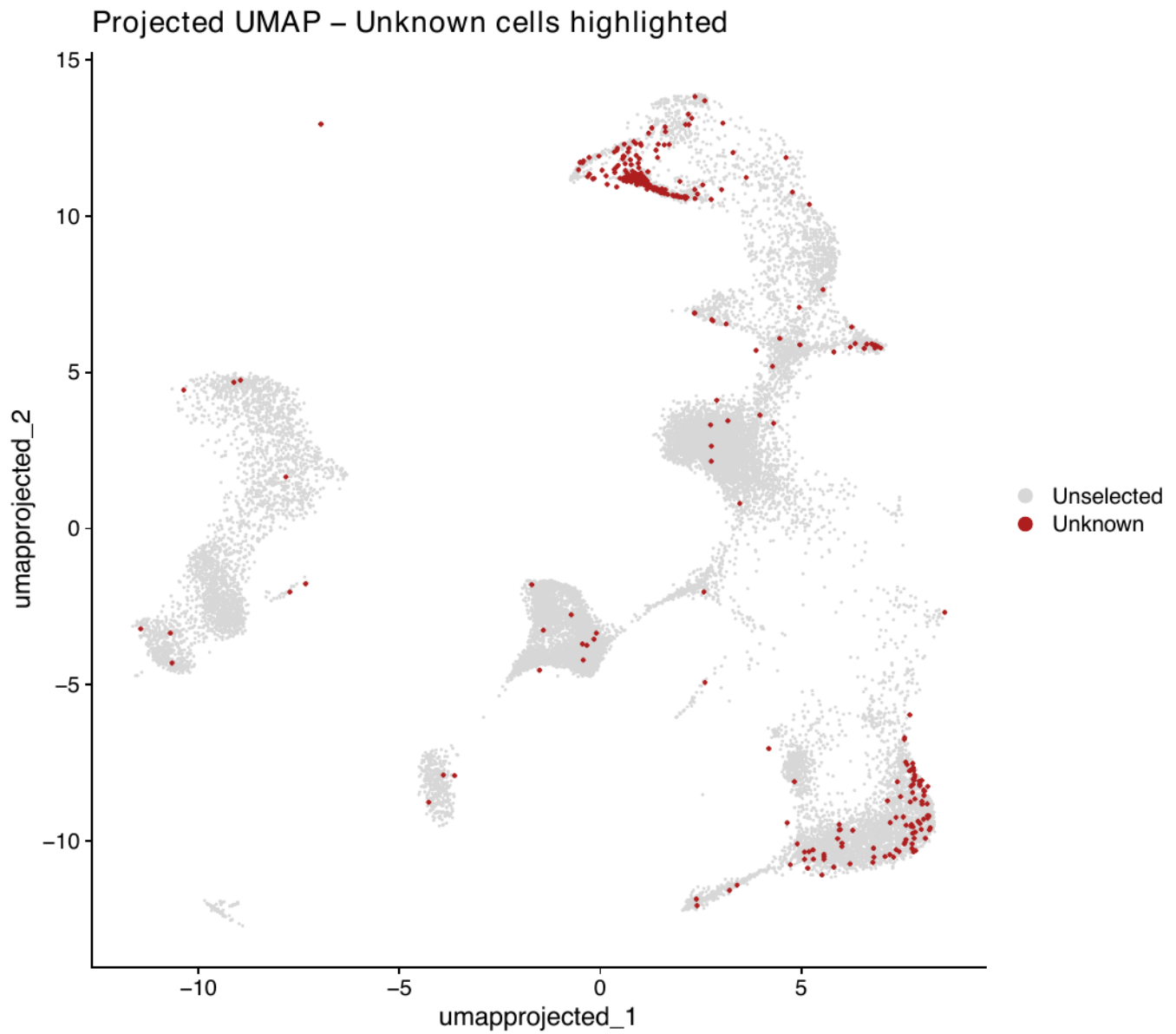
The reference-projected UMAP, same colouring.

boneMarrowMap-projected UMAP – annotation incl. Unknown



03a_umap_projected_unknown_highlighted.pdf

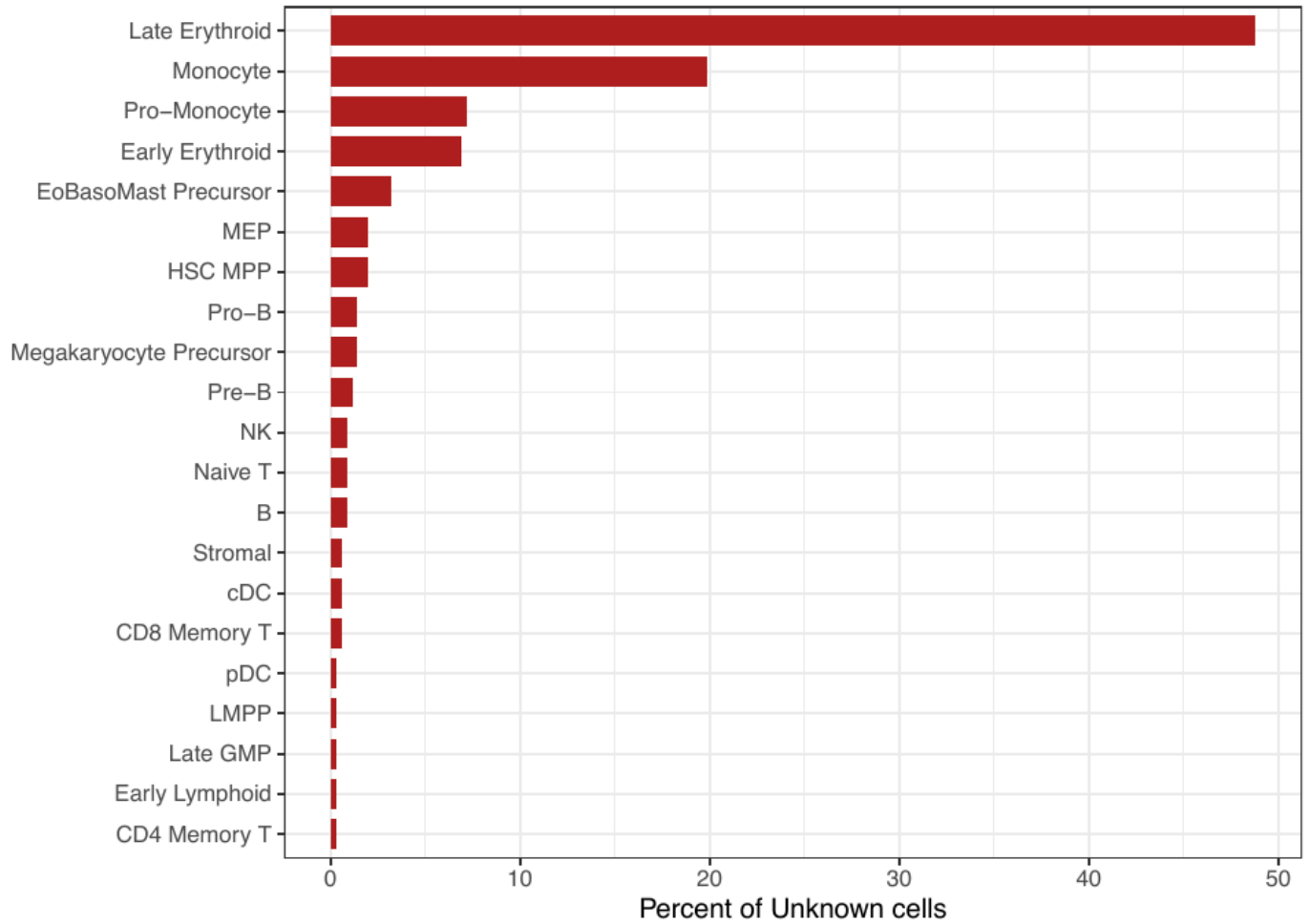
Unknown cells on the projected UMAP.



04_unknown_nearest_neighbor_identity_umap_projected.pdf

The same vote in reference space instead swings to late erythroid. The two embeddings disagree, which is the tell: these cells are being placed by what they lack rather than by what they express, so their apparent identity depends on where you ask.

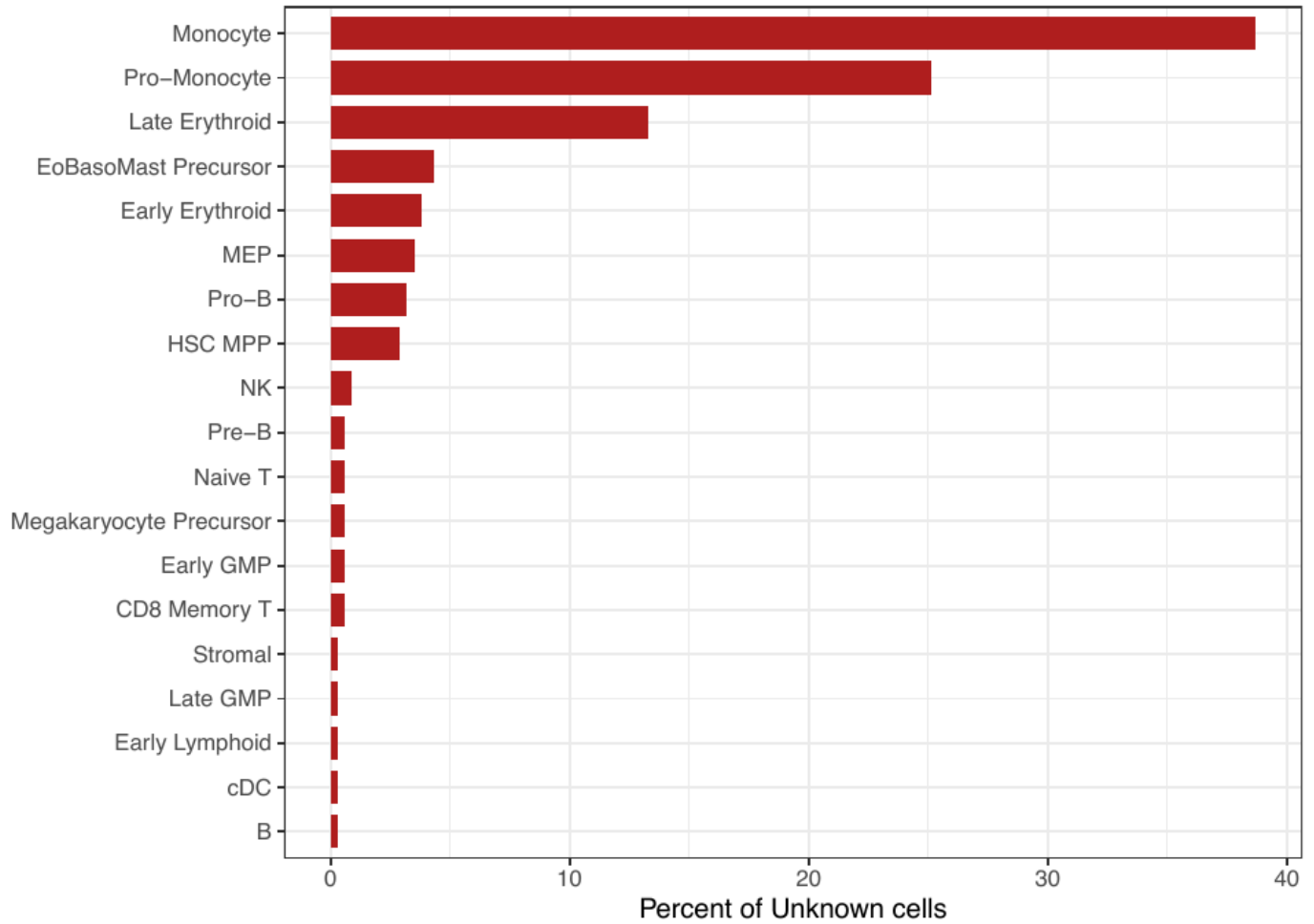
Unknown cells' nearest annotated neighbor (umap_projected, k=15)



04_unknown_nearest_neighbor_identity_umap.pdf

If each Unknown cell is forced to take the majority label of its 15 nearest annotated neighbours in the object's own embedding, they come out mostly monocyte and pro-monocyte.

Unknown cells' nearest annotated neighbor (umap, k=15)



Nearest annotated neighbour, native embedding

nearest_known_celltype	n	percent
Monocyte	134	38.7
Pro-Monocyte	87	25.1
Late Erythroid	46	13.3
EoBasoMast Precursor	15	4.3
Early Erythroid	13	3.8
MEP	12	3.5
Pro-B	11	3.2
HSC MPP	10	2.9

Nearest annotated neighbour, projected embedding

nearest_known_celltype	n	percent
Late Erythroid	169	48.8
Monocyte	69	19.9
Pro-Monocyte	25	7.2
Early Erythroid	24	6.9
EoBasoMast Precursor	11	3.2
HSC MPP	7	2.0

nearest_known_celltype	n	percent
MEP	7	2.0
Megakaryocyte Precursor	5	1.4

4. How much the surviving labels can be trusted

There is only one cutoff in the whole annotation, and it is the mapping-error one. `predict_CellTypes()` calls `knnPredict_Seurat()` with `k = 30`: each query cell is given the majority label of its 30 nearest reference neighbours in harmonised PCA space.

`predicted_CellType_prob` is the fraction of those 30 that voted for the winning label - so it is a vote share, not a posterior probability, and every value it can take is a multiple of 1/30. In LK2 it takes 25 distinct values, the lowest being 0.2 (6 of 30).

`predict_CellTypes()` then does exactly one thing with that number: nothing. It never thresholds on it. The only cell it sets to NA is the cell whose `mapping_error_QC` is Fail. So a label assigned on a 7-to-30 plurality is stored with the same standing as one assigned unanimously, and it is up to whoever reads the object to decide whether to trust it. Nothing downstream in this project currently filters on it either.

That makes the distribution below worth reading properly rather than skimming: it is the only description of how firm each label is.

Label confidence over cells that pass QC

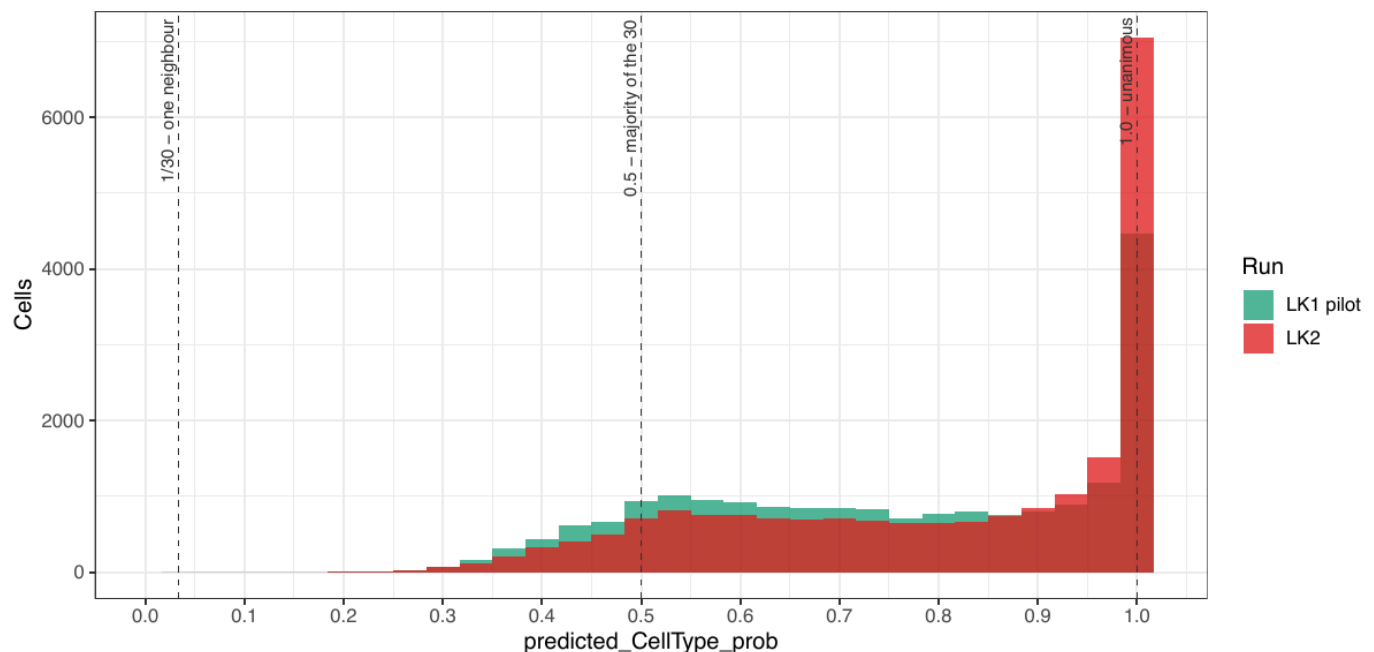
run	n	prob_p05	prob_q25	prob_median	prob_q75	percent_above_0.5	percent_above_0.8
LK1 pilot	19828	0.400	0.567	0.767	0.967	83.8	44.7
LK2	20605	0.433	0.633	0.900	1.000	88.5	57.4

03_label_confidence.pdf

One bar is one neighbour changing its vote. The dashed lines are reading aids, not thresholds the tool applies: 1/30 is the floor, 0.5 is where a label stops being a true majority and becomes a plurality, 1.0 is unanimous. 34% of LK2 labels are unanimous; 8% sit below the majority line and are the ones to be careful with.

Fraction of the `k = 30` nearest reference neighbours voting for the assigned label

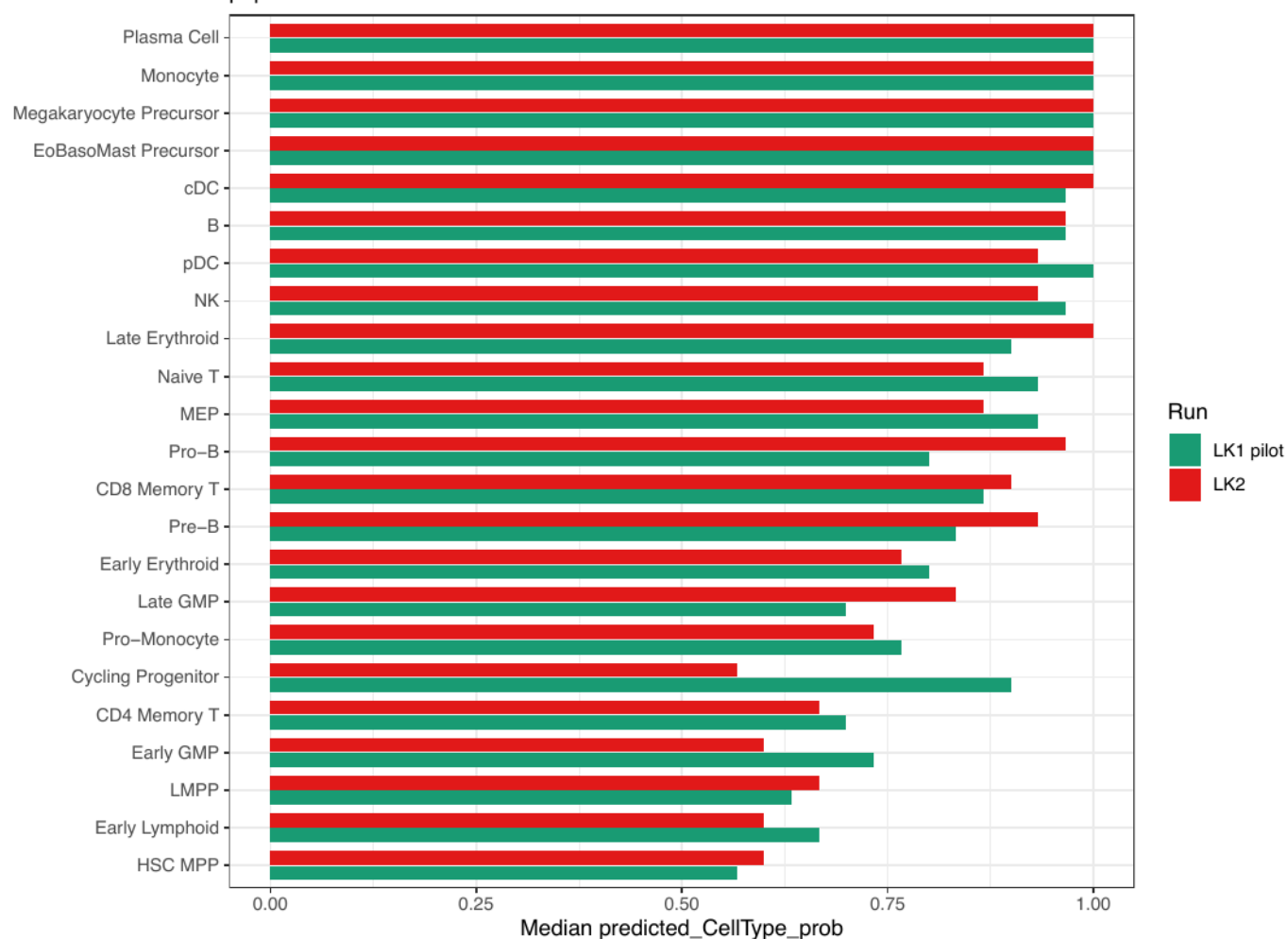
BoneMarrowMap applies no cutoff to this value – the dashed lines are reading aids, not thresholds.
One bar = one neighbour changing its vote.



04_label_confidence_by_celltype.pdf

Median confidence by broad cell type. The low end is HSC MPP, LMPP, Early Lymphoid, Cycling Progenitor and Early GMP; the high end is Monocyte, cDC, pDC, Plasma Cell and Late Erythroid.

Median label confidence by broad cell type
populations with at least 20 cells



This is not a quality gradient. The populations with low confidence are the ones that are genuinely continuous - a cell part way along the HSC-to-LMPP axis should not get a unanimous vote, and the fact that it does not is the reference behaving correctly. It does mean that any analysis that treats HSC MPP as a discrete population is working with a soft boundary, which is worth remembering for the biotin split and for the surfaceome comparisons.

Ten least confident broad cell types (>= 20 cells)

run	predicted_CellType_Broad	n	median_prob
LK1 pilot	HSC MPP	4537	0.567
LK2	Cycling Progenitor	119	0.567
LK2	Early GMP	129	0.600
LK2	Early Lymphoid	245	0.600
LK2	HSC MPP	3857	0.600
LK1 pilot	LMPP	490	0.633
LK1 pilot	Early Lymphoid	394	0.667
LK2	CD4 Memory T	473	0.667
LK2	LMPP	767	0.667
LK1 pilot	CD4 Memory T	806	0.700

Ten most confident

run	predicted_CellType_Broad	n	median_prob
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run	predicted_CellType_Broad	n	median_prob
LK1 pilot	EoBasoMast Precursor	131	1
LK1 pilot	Megakaryocyte Precursor	40	1
LK1 pilot	Monocyte	1886	1
LK1 pilot	Plasma Cell	25	1
LK1 pilot	pDC	779	1
LK2	EoBasoMast Precursor	261	1
LK2	Late Erythroid	498	1
LK2	Megakaryocyte Precursor	117	1
LK2	Monocyte	2868	1
LK2	Plasma Cell	39	1

5. What predict_CellTypes() actually does

initial_CellType versus predicted_CellType

run	n	n_initial_NA	n_final_NA	n_QC_fail	final_NA_equals_QC_fail	percent_label_unchanged
LK1 pilot	21541	0	1713	1713	TRUE	100
LK2	21800	0	1195	1195	TRUE	100

`initial_CellType` is never NA, `predicted_CellType` is NA for exactly the QC-failed cells, and wherever both are present they are identical for 100% of cells in both runs. So the second call is a gate, not a second opinion: it does not revise labels, it withholds them. That matters for anyone reading the object - `initial_CellType` is not a draft that got improved, it is the same answer without the QC applied.

6. Fine versus broad labels

Fine and broad label vocabularies

run	n_fine_labels	n_broad_labels	fine_labels_in_broad_vocab	percent_cells_Other_if_keyed_on_fine	percent_cells_Other_if_keyed_on_broad
LK1 pilot	53	24	9	84.3	0
LK2	53	24	9	89.1	0

This is exactly what happened to the pilot's `10_lineage_composition_by_sampleID.csv` and `12_specific_celltype_composition_by_sampleID.csv` - the latter has an all-NA cell type column on disk. Step 01 here is keyed on the broad column, and step 02 (the LK2 version that became the pipeline process) already was.

7. The mapping-error threshold

2.5 is the BoneMarrowMap default, and it is what the standalone scripts in steps 01 and 02 ran. **The pipeline does not use it.**

`nextflow/bin/annotate.R` runs at `MAD_threshold = 4`; only `nextflow/bin/lib_early_qc_annotation.R` still has 2.5. So the figures in `results/seurat_annotated/` are at 2.5 and anything coming out of `results_nf/` is at 4, and the two are not interchangeable.

The score itself does not depend on the threshold - `calculate_MappingError()` stores `mapping_error_score` per cell and then calls `Pass/Fail` on `score < median(score) + MAD_threshold * mad(score)`, with median and MAD taken over the whole object. So the sweep below is computed directly from the stored score rather than by re-projecting. Recomputing the 2.5 call reproduces the stored `mapping_error_QC` for 100% of cells in both runs, which is what makes the other two columns trustworthy.

Rejection rate by run, at `MAD_threshold` 2.5, 3 and 4

run	MAD_threshold	n_cells	n_rejected	percent_rejected
LK1 pilot	2.5	21541	1713	7.95
LK1 pilot	3.0	21541	1369	6.36

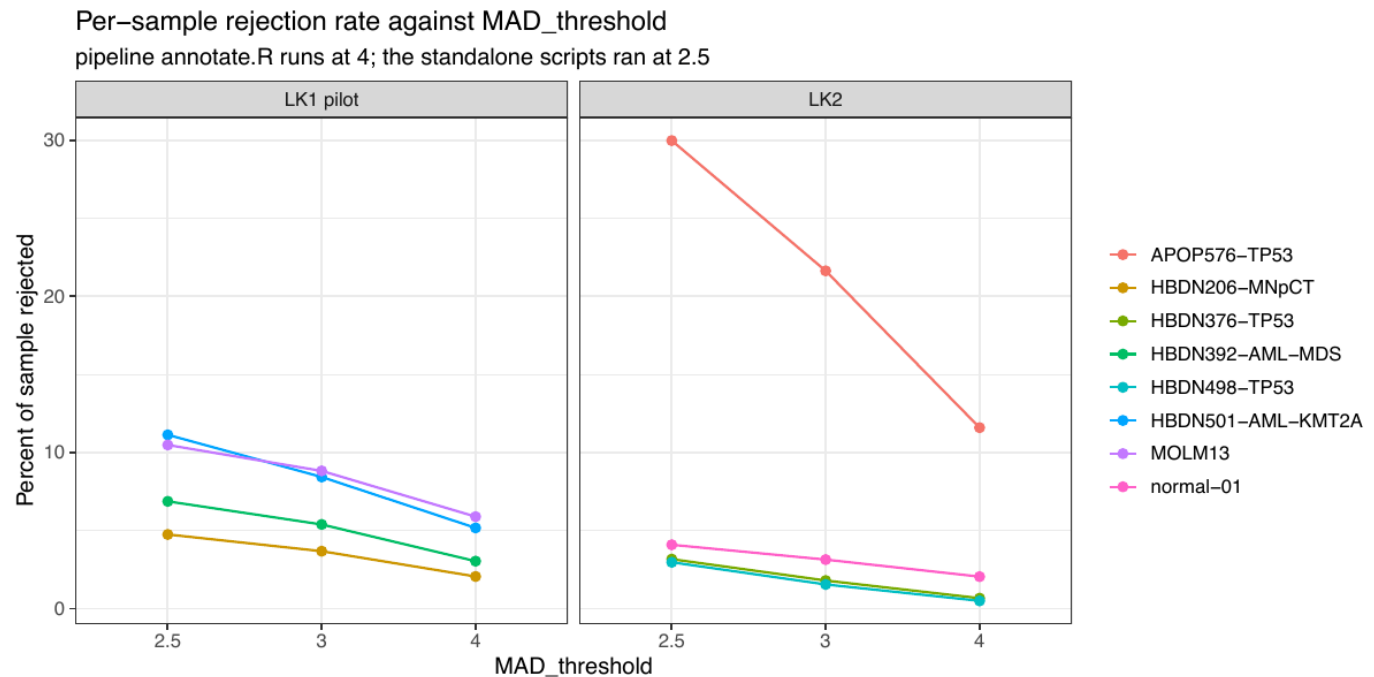
run	MAD_threshold	n_cells	n_rejected	percent_rejected
LK1 pilot	4.0	21541	842	3.91
LK2	2.5	21800	1195	5.48
LK2	3.0	21800	781	3.58
LK2	4.0	21800	381	1.75

Rejection rate per sample, worst first (percent of that sample's cells)

run	sample	MAD 2.5	MAD 3	MAD 4
LK2	APOP576-TP53	29.97	21.63	11.59
LK1 pilot	HBDN501-AML-KMT2A	11.13	8.43	5.17
LK1 pilot	MOLM13	10.48	8.82	5.89
LK1 pilot	HBDN392-AML-MDS	6.87	5.39	3.04
LK1 pilot	HBDN206-MNpCT	4.75	3.68	2.06
LK2	normal-01	4.09	3.14	2.05
LK2	HBDN376-TP53	3.17	1.80	0.67
LK2	HBDN498-TP53	2.97	1.55	0.50

06_MAD_threshold_sweep.pdf

Every sample moves in the same direction and keeps its rank - the threshold changes how much is cut, not which samples are hardest to place. APOP576-TP53 still loses 11.6% at 4, more than any other sample loses at 2.5.



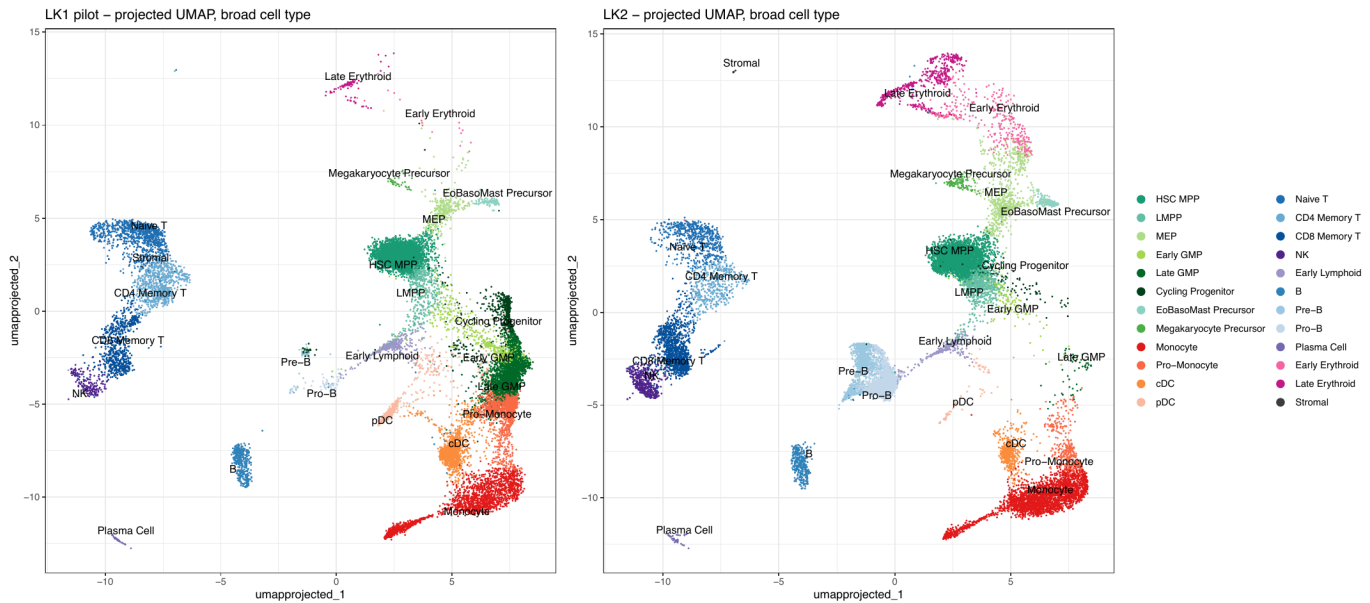
Going from 2.5 to 4 roughly halves the rejection rate - 7.95% to 3.91% in the pilot, 5.48% to 1.75% in LK2 - and 3 sits between them. Since section 3 shows the rejected cells are shallow, ribosome-poor and enriched for HTO negatives, the cells recovered by relaxing the threshold are mostly ones there is no good reason to keep. That argues for 2.5 rather than the 4 the pipeline currently runs, but it has not been changed, so the discrepancy stands and is worth resolving deliberately rather than by drift.

8. Projection output

The projected UMAP for both runs, coloured by `predicted_CellType_Broad` in the project palette. The versions in `results/seurat_annotated/` are coloured by the 53-level fine label in ggplot defaults and exist for one run at a time, which is why step 04a redraws them.

07_projected_umap_broad_both_runs.pdf

LK1 pilot and LK2 side by side. Same reference space, same palette, so the compositional difference between the two runs can be read directly off the map.



9. What was run

Step	Script	What
00	00.celltype_palette.R	The bespoke broad-cell-type palette, lineage vocabulary and the check_palette() guard, defined once and sourced by steps 01-03. Run directly, it redraws the reference UMAPs.
01	01.annotate_LK1_pilot.R	BoneMarrowMap projection of the LK1 pilot, MAD 2.5.
02	02.annotate_LK2.R	The same projection on LK2. This is the version that became the pipeline's 08_annotate process.
03	03.check_unannotated_cells.R	Re-projects the LK2 pipeline checkpoint at MAD 4 and characterises the cells that still fail - QC covariates, donor, HTO class, native cluster, nearest annotated neighbour.
04	04.annotation_benchmark_summary.R	Scores the annotation from the metadata tables alone: reject rates, rejected-cell profile, label confidence, initial-versus-final label, fine-versus-broad vocabulary, the MAD threshold sweep.
04a	04a.projected_umaps.R	Redraws the projected UMAP for both runs in the broad palette. Separate from step 04 because it is the only step that loads the ~450 MB projected objects.
05	05.celltype_annotation_report.Rmd	This report.

Steps 01 and 02 are collated from backup/10.annotate.R, backup/10a.annotate.R, backup/10b.annotate_DSB.R and 8.annotate.R. 10.annotate.R is superseded - none of its output names survive in results/. 10a and 10b are the same script run twice with different ADT assays, which is now the step 01 argument. Three fixes went in on the way:

- 10a had a marker_map / plot_df block sitting above the code that creates the objects it references, so it could only ever have been run interactively out of order. Removed.
- 10a.heatmap_cell_type_all.pdf is 0 bytes on disk because scaling a zero-variance row gives NaN and pheatmap will not draw it. Fixed in both steps.
- the numbat export loops at the tails of 10a and 8.annotate.R belong to the CNV work, not to annotation. Dropped from both.
- every script carried its own copy of the palette, and the copies had drifted. All of them listed GMP - a label the reference never emits, since it splits that population into Early GMP and Late GMP - and none had an entry for Stromal, which the reference does emit. Because the plots do factor(CellType_Broad, levels = names(celltype_cols)), every stromal cell was being set to NA and dropped at the plotting step. That is why stromal cells looked unannotated: they were annotated, they were being discarded when drawn. The palette now lives in step 00 with Stromal in grey, and check_palette() raises a warning instead of silently dropping anything it does not recognise.

10. Data locations

Tools

What	Where
BoneMarrowMap reference + uwot model	annotation/BoneMarrowMap_SymphonyReference.rds, annotation/BoneMarrowMap_uwot_model.uwot
Pilot input (CITE/DSB object)	results/cite_qc_then_dsb/260423_VH01624_453_222HWMYNNX/LK1-GEX/seurat_demux_ADT_CLR_DSB_QC.rds
LK2 input (CITE/DSB object)	results/cite_seq_dsb/260528_VH01624_464_222K7VKNNX/LK2-GEX/seurat_demux_ADT_CLR_DSB_QC.rds
LK2 pipeline checkpoint (step 03 input)	results_nf/260528_VH01624_464_222K7VKNNX/rds/07_cite_qc/LK2-GEX/rds/seurat_demux_ADT_CLR_DSB_QC.rds
Pilot annotation outputs	results/seurat_annotated/260423_VH01624_453_222HWMYNNX
LK2 annotation outputs	results/seurat_annotated/260528_VH01624_464_222K7VKNNX
Unannotated-cell investigation	results/seurat_annotated/260528_VH01624_464_222K7VKNNX/LK2-GEX_results_nf_unannotated_check
Benchmark tables and figures (step 04)	results/benchmarking/celltype_annotation

Tool	Version	Use
BoneMarrowMap	0.2.0	Zeng et al. normal bone marrow Symphony reference and helpers
symphony	0.1.2	reference mapping
uwot	0.2.4	UMAP projection into reference coordinates
Seurat	5.4.0	object handling, DimPlot/FeaturePlot
class	7.3.23	kNN vote for the Unknown-cell identity assignment
pheatmap	1.0.13	composition heatmaps

11. Caveats

- steps 01 and 02 are the collated, cleaned versions of the scripts that produced the original annotation, and were not re-executed - the projections they wrote in May and June are what everything here is computed from. Steps 00, 03, 04 and 04a were run fresh, which is what `Analysis finished` reflects. Step 03 reused its saved annotated object rather than re-projecting, so only its figures are new, not its numbers
- the pilot and LK2 numbers come from two different input objects and two different upstream CITE QC runs, so the difference in reject rate between runs is not a controlled comparison. The per-sample ordering within each run is, and so is the threshold sweep, which moves the same cells against the same score
- step 03 runs on the pipeline checkpoint at `MAD_threshold = 4`, not on the step 02 object at 2.5. Its 1.35% Unknown and step 02's 5.48% Fail are therefore both correct and not directly comparable - the sweep in section 7 is the like-for-like version, since it recomputes both thresholds against the same cells
- the ADT/CITE marker projections that steps 01 and 02 also write are not shown here. The step 01 `ADT_ASSAY` argument still selects between `ADT_CLR` and `CITE_DSB` for them, and both were run; the figures are on disk under `projectionFigures/`
- APOP576-TP53's 30% reject rate has not been chased down beyond noting it. It is the sample to look at first if that number matters
- pseudotime is inherited from the reference, not computed on our cells, so it should be read as "where in the normal trajectory does this cell project", not as a trajectory of the leukaemia